A SOLUTION CONTAINING A PURELY INSEPARABLE DEGREE P STEP CAN BE REPLACED BY A SEPARABLE SOLUTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author writes 'Some details omitted.' as the last sentence of the counted proof span; that sentence is kept verbatim and marks the step left to the reader, namely the verification of the two formal smoothness conditions on the pi^2-quotients. The references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Solutions for an extension of discrete valuation rings and the construction of the rings A_1, B_1, L_1. assumption (4) at statement line 35783 and used throughout the counted proof span. Supplies the hypothesis that is deformed into a separable solution.
- Purely inseparable extensions of degree p in characteristic p and their description by x^p - a. assumption (5) at statement line 35784 and used at proof lines 35809-35813. Gives the element alpha and the constant a_1 that the deformation replaces.
- Nagata rings and finiteness of integral closures. assumption (3) at statement line 35782 and used at proof lines 35795-35803. Makes the auxiliary integral closures finite.
- The extension-construction lemma and its functoriality, applied at f = pi^2. invoked at proof lines 35820-35822 and 35849-35852. Produces the integer n and the extensions A_3 and B_3 with the required isomorphism of quotients.
- The criterion for formal smoothness of an extension of discrete valuation rings. invoked at proof lines 35880-35884. Is the final criterion that certifies K_3/K_1 as a solution.

Source reference. Original lemma 030U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3961–3961. Cross-chapter navigation pointer into

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the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 09HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12006–12006. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 35691: sources/stacks-more-algebra/pinned-original-source.tex, source lines 35691–35691. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 32989: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32989–32989. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 35667: sources/stacks-more-algebra/pinned-original-source.tex, source lines 35667–35667. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 33882: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33882–33882. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 35728: sources/stacks-more-algebra/pinned-original-source.tex, source lines 35728–35728. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let p be a prime number. Let $A \subset B$ be an extension of discrete valuation rings with fraction field extension L/K. Let $K_2/K_1/K$ be a tower of finite field extensions. Assume*

(1) *K has characteristic p,*
(2) *L/K is separable,*
(3) *B is Nagata,*
(4) *K_2 is a solution for $A \subset B$,*
(5) *K_2/K_1 is purely inseparable of degree p.*

Then there exists a separable extension K_3/K_1 which is a solution for $A \subset B$.

Proof. Let us use notation as in Remark 33882; we will use all the observations made there. Since L/K is separable, the algebra $L_1 = L \otimes_K K_1$ is reduced (Algebra, Lemma 030U). Since B is Nagata, the ring extension $B \subset B_1$ is finite where B_1 is the integral closure of B in L_1 and B_1 is a Nagata ring. Similarly, the ring A is Nagata by Lemma 35667 hence $A \subset A_1$ is finite and A_1 is a Nagata ring too. Moreover, the same assertions are true for K_2, i.e., $L_2 = L \otimes_K K_2$ is reduced, the ring extensions $A_1 \subset A_2$ and $B_1 \subset B_2$ are finite where A_2, resp. B_2 is the integral closure of A, resp. B in K_2, resp. L_2.

Let $\pi \in A$ be a uniformizer. Observe that π is a nonzerodivisor on K_1, K_2, A_1, A_2, L_1, L_2, B_1, and B_2 and we have $K_1 = (A_1)_\pi$, $K_2 = (A_2)_\pi$, $L_1 = (B_1)_\pi$, and $L_2 = (B_2)_\pi$. We may write $K_2 = K_1(\alpha)$ where $\alpha^p = a_1 \in K_1$, see Fields, Lemma 09HI. After multiplying α by a power of π we may and do assume $a_1 \in A_1$. For the rest of

the proof it is convenient to write $K_2 = K_1[x]/(x^p - a_1)$ and $L_2 = L_1[x]/(x^p - a_1)$. Consider the extensions of rings

$$A'_2 = A_1[x]/(x^p - a_1) \subset A_2 \quad \text{and} \quad B'_2 = B_1[x]/(x^p - a_1) \subset B_2$$

We may apply Lemma 35691 to $A'_2 \subset A_2$ and $f = \pi^2$ and to $B'_2 \subset B_2$ and $f = \pi^2$. Choose an integer n large enough which works for both of these.

Consider the algebras

$$K_3 = K_1[x]/(x^p - \pi^{2n}x - a_1) \quad \text{and} \quad L_3 = L_1[x]/(x^p - \pi^{2n}x - a_1)$$

Observe that K_3/K_1 and L_3/L_1 are finite étale algebra extensions of degree p. Consider the subrings

$$A'_3 = A_1[x]/(x^p - \pi^n x - a_1) \quad \text{and} \quad B'_3 = B_1[x]/(x^p - \pi^n x - a_1)$$

of $K_3 = (A'_2)_\pi$ and $L_3 = (B'_3)_\pi$. We are going to construct a commutative diagram

$$\begin{array}{ccc} B'_2/\pi^{2n}B'_2 & \xrightarrow[\psi']{} & B'_3/\pi^{2n}B'_3 \\ \uparrow & & \uparrow \\ A'_2/\pi^{2n}A'_2 & \xrightarrow{\varphi'} & A'_3/\pi^{2n}A'_3 \end{array}$$

Namely, φ' is the unique A_1-algebra isomorphism sending the class of x to the class of x. Similarly, ψ' is the unique B_1-algebra isomorphism sending the class of x to the class of x. By our choice of n we obtain, via Lemma 35691 and Remark 35728 finite ring extensions $A'_3 \subset A_3$ and $B'_3 \subset B_3$ such that $A'_3 \to B'_3$ extends to a ring map $A_3 \to B_3$ and a commutative diagram

$$\begin{array}{ccc} B_2/\pi^2 B_2 & \xrightarrow[\psi]{} & B_3/\pi^2 B_3 \\ \uparrow & & \uparrow \\ A_2/\pi^2 A_2 & \xrightarrow{\varphi} & A_3/\pi^2 A_3 \end{array}$$

with all the properties asserted in the references mentioned above (in particular φ and ψ are isomorphisms).

With all of this data in hand, we can finish the proof. Namely, we first observe that A_3 and B_3 are finite products of Dedekind domains with π contained in all of the maximal ideals. Namely, if $\mathfrak{p} \subset A_3$ is a maximal ideal, then $\pi \in \mathfrak{p}$ as $A \to A_3$ is finite. Then $\mathfrak{p}/\pi^2 A_3$ corresponds via φ to a maximal ideal in $A_2/\pi^2 A_2$ which is principal as A_2 is a finite product of Dedekind domains. We conclude that $\mathfrak{p}/\pi^2 A_3$ is principal and hence by Nakayama we see that $\mathfrak{p}(A_3)_\mathfrak{p}$ is principal. The same argument works for B_3. We conclude that A_3 is the integral closure of A in K_3 and that B_3 is the integral closure of B in L_3. Let $\mathfrak{q} \subset B_3$ be a maximal ideal lying over $\mathfrak{p} \subset A_3$. To finish the proof we have to show that $(A_3)_\mathfrak{p} \to (B_3)_\mathfrak{q}$ is formally smooth in the $\mathfrak{q}$-adic topology. By the criterion of Lemma 32989 it suffices to show that $\mathfrak{p}(B_3)_\mathfrak{q} = \mathfrak{q}(B_3)_\mathfrak{q}$ and that the field extension $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is separable. This is true because we may check both assertions by looking at the ring map $A_3/\pi^2 A_3 \to B_3/\pi^2 B_3$ and this is isomorphic to the ring map $A_2/\pi^2 A_2 \to B_2/\pi^2 B_2$ where the corresponding statement holds by our assumption that K_2 is a solution for $A \subset B$. Some details omitted. $\square$